

The real weighted box integral: mass, recursion, permutations

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Abstract

Third step of the mixed-ratio general- d monomial programme (Astra #15, route A). The real weighted box integral $W_\ell(N) = \int_{(0,1]^d} \prod_i t_i^{\ell_i-1} e^{-\beta N \prod_i t_i} dt$ is defined as the `toReal` of the $[0, \infty]$ -valued `weightedBoxIntegral`, and the four facts the mixed-ratio induction needs are proved: the mass bound $\int_{(0,1]^d} \prod_i t_i^{w_i} dt = \prod_i 1/(w_i + 1)$ (hence $0 \leq W_\ell(N) \leq \prod_i 1/\ell_i$ for $\beta N \geq 0$); measurability of $N \mapsto W_\ell(N)$; the real recursion $W_\ell(N) = \int_0^1 t^{\ell_0-1} W_{\ell'}(Nt) dt$ with ℓ' the tail; and permutation invariance of the weighted box integral. Zero `sorry/axiom`.

1 The things formalised

```
theorem weightedBoxIntegral_one : d w, ( i, -1 < w i) →
  weightedBoxIntegral d w (fun _ => 1) = ofReal ( i, 1 / (w i + 1))
theorem weightedBoxIntegral_expKernel_lt_top (d w) (hw) ( N) (hN : 0 ≤ N) :
  weightedBoxIntegral d w (expKernel N) <
  ofReal ( i, 1 / (w i + 1))
def mixedBoxReal (d) ( : Fin d → ) ( N : ) : :=
  (weightedBoxIntegral d (fun i => i - 1) (expKernel N)).toReal
theorem mixedBoxReal_le (d) (h : i, 0 < i) ( N) (hN : 0 ≤ N) :
  mixedBoxReal d N i, 1 / i ≤ ofReal ( i, 1 / (w i + 1))
theorem measurable_mixedBoxReal (d) : Measurable fun N => mixedBoxReal d N
theorem mixedBoxReal_succ (d) ( : Fin (d+1) → ) (h) ( N) (h : 0 < ) (hN : 0 < N) :
  mixedBoxReal (d + 1) N
  = t in Ioc 0 1, t ^ ( 0 - 1) * mixedBoxReal d (Fin.tail ) (N * t)
theorem weightedBoxIntegral_comp_perm (d w g) ( : Equiv.Perm (Fin d)) :
  weightedBoxIntegral d (w ) g = weightedBoxIntegral d w g
```

2 Mathematics

The mass bound is the constant-kernel case of the weighted recursion, $\int_0^1 a^{w_0} da = 1/(w_0 + 1)$ at each level; since $e^{-\beta N z} \leq 1$ on the box for $\beta N \geq 0$, the exponential integral is finite and bounded by the mass. The real recursion is the $[0, \infty]$ recursion of unit 174 after `integral_toReal`, which needs the parametric integral $M \mapsto W(M)$ to be measurable (a Tonelli-type statement, `Measurable.lintegral_prod_right'`) and finite almost everywhere on $(0, 1]$. Permutation invariance transports the integral along `MeasurableEquiv.piCongrLeft`, which preserves the product of restricted Lebesgue measures, and reindexes the two products with `Equiv.prod_comp`.

3 Paper-facing meaning

W_ℓ is the substituted normal moment integral with exponents $\ell_i = (h_i + 1)/(2k_i)$; the recursion peels one normal coordinate, and permutation invariance lets the induction of the next unit always

peel a coordinate that does *not* realise the minimal ratio, which is where the dominated-coordinate transfer lemma applies.

4 Lean notes

- `positivity` cannot prove $0 \leq 1/(w + 1)$ from a hypothesis $-1 < w$; use `div_nonneg zero_le_one` (by `linarith`).
- `lintegral_mul_const` with a composed measurability proof leaves a `(ofReal f)` a pattern that does not match the goal; `lintegral_mul_const'` with `ENNReal.ofReal_ne_top` avoids measurability altogether.
- `mul_le_mul_left'` is gone on this pin; `gcongr` handles $a \cdot b \leq a \cdot c$ in 0ω .
- `Equiv.piCongrLeft_apply_apply` takes the family and the equivalence explicitly: `Equiv.piCongrLeft_apply_apply (fun _ => ) y b`.
- `rw [restrict_unitBox]` rewrites both occurrences at once; a second `rw` fails.
- Rewrite the measure-preserving transport on the right-hand side with `conv_rhs` so that the pointwise goal has the composed integrand on the side where the reindexing lemma applies.